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SUMMARY

Lead Software Engineer with **8+ years** in production distributed systems and data platforms — now building Agentic Workflows. Led **agentic SDLC adoption across 16+ engineers**, hands-on with Claude Code and Cursor; **building agentic playbooks with MCP-tool-use and curating offline eval sets from production datasets**. Takes end-to-end ownership for product delivery — from collaboration with Product Managers through shipping to production.

AI & AGENTIC ENGINEERING

- Building **Gavel** — an open-source MCP evaluation harness for agent-built data pipelines. Features declarative YAML sandboxes for system components, golden vs. candidate datasets and the metric rubric; a **6-tool MCP stdio server** consumed by Cursor and Claude Code returns deterministic pass/fail verdicts. Live adapters (Postgres, Dagster, orchestrator) catch **silent success**. Tech Stack: Python, Pydantic, MCP SDK, Shapely.
- **Agentic SDLC, end to end**: built and operated LLM-agent workflows automating Picarro's full SDLC — from product grooming with PMs (human-in-the-loop) through deployment — for the next-gen Geospatial Analytics platform. **Lifted sprint velocity 1.7x–2.3x** vs. a comparable ~1-month build.
- **Context grounding via MCP**: connected agents to Architecture Review Documents, decision trails, runbooks and live systems (Confluence, Jira, Postgres/PostGIS, Jenkins, Docker, SSH) accelerating incremental feature development.
- **Agentic Data Validation Playbook**: Built validation-on-loop across three deployment environments (local, staging, integration) — triggering Kafka/Dagster orchestration jobs, running DB parity checks and generating map-based visualizations — **~4x faster validation cycles** vs. manual runbook execution.
- Curated **Eval harness on production golden datasets** for an agent-built geospatial processing pipeline; ran **failure-mode analysis** separating genuine defects from algorithmic drift and fed results back to self-correct agent-generated code.
- **Agent-built production app (0→1 in 8 hours) using Cursor**: solo-shipped a self-service portal for data transfer visibility between Picarro and Customer ERP systems. Cuts a **4-hour manual workflow to seconds**. Used by Product Managers. Tech Stack: React, FastAPI, Keycloak Auth, Postgres.

SKILLS

AI & Agentic Engineering: LLM agents, agentic workflows, agent orchestration, MCP Server, context engineering, human-in-the-loop design, Claude Code, Cursor, Anthropic API, LangGraph, LangChain

Agent Evaluation & Observability: eval harness design, offline & regression evals, golden datasets, deterministic metrics, failure-mode analysis, structured evidence reporting

Backend & Distributed Systems: Python (FastAPI, Pydantic, SQLAlchemy, Flask, Asyncio), microservices, event-driven architecture, distributed systems

Data & Pipelines: PostgreSQL, MS SQL Server, DynamoDB, Apache Kafka, Apache Iceberg, Dagster

Identity & Auth: OAuth 2.0, OpenID Connect, SSO, RBAC architectures, Keycloak

Cloud & Infrastructure: AWS (EKS, EC2, IAM, Lambda, RDS, S3, Glue), Kubernetes, Docker, DevContainers

DevOps & CI/CD: Jenkins, ArgoCD, GitHub Actions, Artifactory / JFrog, Pipeline-as-Code

EXPERIENCE

Lead Software Engineer | Picarro Inc. | Santa Clara, CA

April 2026 – Present

- Leading **AI-augmented SDLC and agentic workflow adoption across 16+ engineers** — establishing the evaluation cadence for agent output, prompt patterns and reusable playbooks that raise engineering velocity.
- Drove architecture for an event-driven geospatial pipeline with Product, SRE and QA; analyzed **724K production surveys** showing 12.9x peak-to-average workload variance, justifying a Kafka worker pattern (**85% resource efficiency vs. 11% always-on**). Adopted as the reusable standard for Auto-compute pipelines.

Senior Software Engineer | Picarro Inc. | Santa Clara, CA

April 2022 – March 2026

- **Mobile app program rescue (OneGas):** tech-led a cross-functional team of 7 after mid-stream disruption; restructured execution, restored stakeholder confidence in 6 weeks, delivered to production in 6 months. **Reduced FastAPI endpoints response time by 10× (10s → 1s)** via SQLAlchemy query redesign.
- Architected production **OAuth 2.0 based RBAC framework** for Picarro's mobile app. Prepared Role matrix, scope definitions, policy generation and permissions evaluation. Implemented Authorization Code Flow with PKCE and FastAPI auth middleware — securely onboarding **39 customers (~10K users)**.
- **Data migration for GDPR compliance:** solo-built an autonomous API-driven fleet monitoring and switchover portal (Tailscale P2P mesh, FastAPI control plane, React) enabling **zero-downtime migration of 80+ analyzers** across 7 European customers.
- Led MS SQL Server → AWS RDS migration for the production survey DB with zero downtime; S3 backup design **cut the backup window 15 → 5 hours** and enabled daily DR backups.
- **Dev Tooling & Productivity improvement:** Established **CI/CD Pipeline-as-Code** for Build versioning, introduced DevContainers **accelerating developer onboarding from 1-2 days to 15 minutes**. Pioneered **branch-aware semantic versioning** for end-to-end traceability across build artifacts, Jira and GitHub — cutting unclear-build-version tickets from 5–6 per release cycle to **zero**.

Software Engineer | Picarro Inc. | Santa Clara, CA

September 2019 – March 2022

- **Reduced dashboard latency ~100× (14s → 140ms)** via server-side rendering and query optimization (C# / LINQ expressions) — a direct customer-investigation UX improvement.
- Built a Python **OTA fleet upgrader** with .NET Web API for a 165-car customer fleet: async downloads, SHA-256 verification, recovery mechanisms — **100% upgrade success rate in production for 6+ years**.

Software Developer | The Finders | Phoenix, AZ

February 2019 – May 2019

- Built C# automation for Couchbase index alignment and Azure CI/CD for multi-environment deployment; .NET, TDD with XUnit / Moq.

Software Engineer | Tech Mahindra (Client: Applied Materials) | Bengaluru, India

December 2015 – May 2017

- Built .NET MVC modules for a Change Order Request system (C#, Kendo UI, MS SQL Server, Entity Framework); optimized database performance **50% faster queries** via index design, view materialization and stored procedures.

EDUCATION

Master of Science in Computer Science | Arizona State University, Tempe, AZ | GPA: 3.75 | August 2017 – May 2019

Bachelor of Technology in Information Technology | GGS Indraprastha University, Delhi | CPI: 74.72% | August 2011 – May 2015